

Question

The test solution with volume V_1 , containing a certain concentration of peroxodisulfate ions, is acidified with sulfuric acid, and then transferred into an excess volume V_2 of a standard sodium arsenite solution with concentration c_1 . The mixture is boiled. After adding potassium bromide solution and letting it stand for a while, the solution is titrated with a standard potassium bromate solution of concentration c_2 until the solution turns yellow, consuming a volume V_3 of the bromate solution. Subsequently, the same standard sodium arsenite solution is used to titrate the solution until it becomes colorless, consuming a volume V_4 . Please select the correct option. Note that when balancing the equations, the coefficients should be in the simplest integer ratio.

- A.** During the titration process, $\text{S}_2\text{O}_8^{2-}$ can react with AsO_2^- under acidic conditions to produce H_3AsO_4 . This equation requires 3 water molecules on the left side to be balanced.
- B.** During the titration process, H_3AsO_3 reacts with BrO_3^- , and the sum of the coefficients on the right side of the balanced equation is 1 more than the sum of the coefficients on the left side.
- C.** During the titration process, BrO_3^- and Br^- react under acidic conditions, with the sum of coefficients on the left side of the equation being 11 and the sum on the right side being 6.
- D.** During the titration process, after adding the sodium arsenite standard solution again, AsO_2^- reacts with an elemental substance produced in the reaction of BrO_3^- added in the previous step. The sum of the coefficients on both sides of the balanced reaction equation is equal.
- E.** The expression for the sulfate ion concentration is
- $$c = [c_1(V_2 + V_4) - 2c_2V_3]/V_1$$
- F.** Acidification with hydrochloric acid can also be performed before titration.

Answer

Correct Answer: **D**

Detailed Explanation

Under acidic conditions, persulfate ions and arsenite ions react, with the reaction equation as follows:



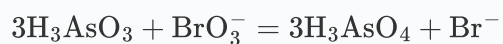
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Equation 1



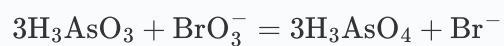
Upon adding bromate ions, arsenous acid and bromate ions undergo a redox reaction, described by the equation:



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Equation 2



Excess bromate ions and bromide ions undergo a comproportionation reaction to form Br_2 , turning the solution yellow, as shown in the equation:



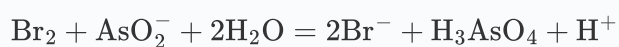
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Equation 3



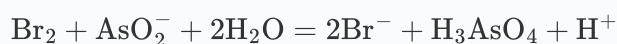
After adding sodium arsenite standard solution, arsenite ions react with Br_2 , causing the solution to become colorless, as represented by the equation:



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Equation 4



Arsenous acid reacts with persulfate and bromate ions. The stoichiometric ratio of arsenous acid to persulfate is 1:1,

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The stoichiometric ratio of arsenous acid to persulfate is 1:1

and the stoichiometric ratio of arsenous acid to bromate is 3:1.

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The stoichiometric ratio of arsenous acid to bromate is 3:1

The total amount of arsenous acid involved in the reaction minus the amount reacting with bromate yields the remaining amount reacting with persulfate. Knowing the volume of the persulfate-containing solution allows determination of the persulfate ion concentration:

$$c = [c_1(V_2 + V_4) - 3c_2V_3]/V_1$$

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Persulfate ion concentration $c = [c_1(V_2 + V_4) - 3c_2V_3]/V_1$